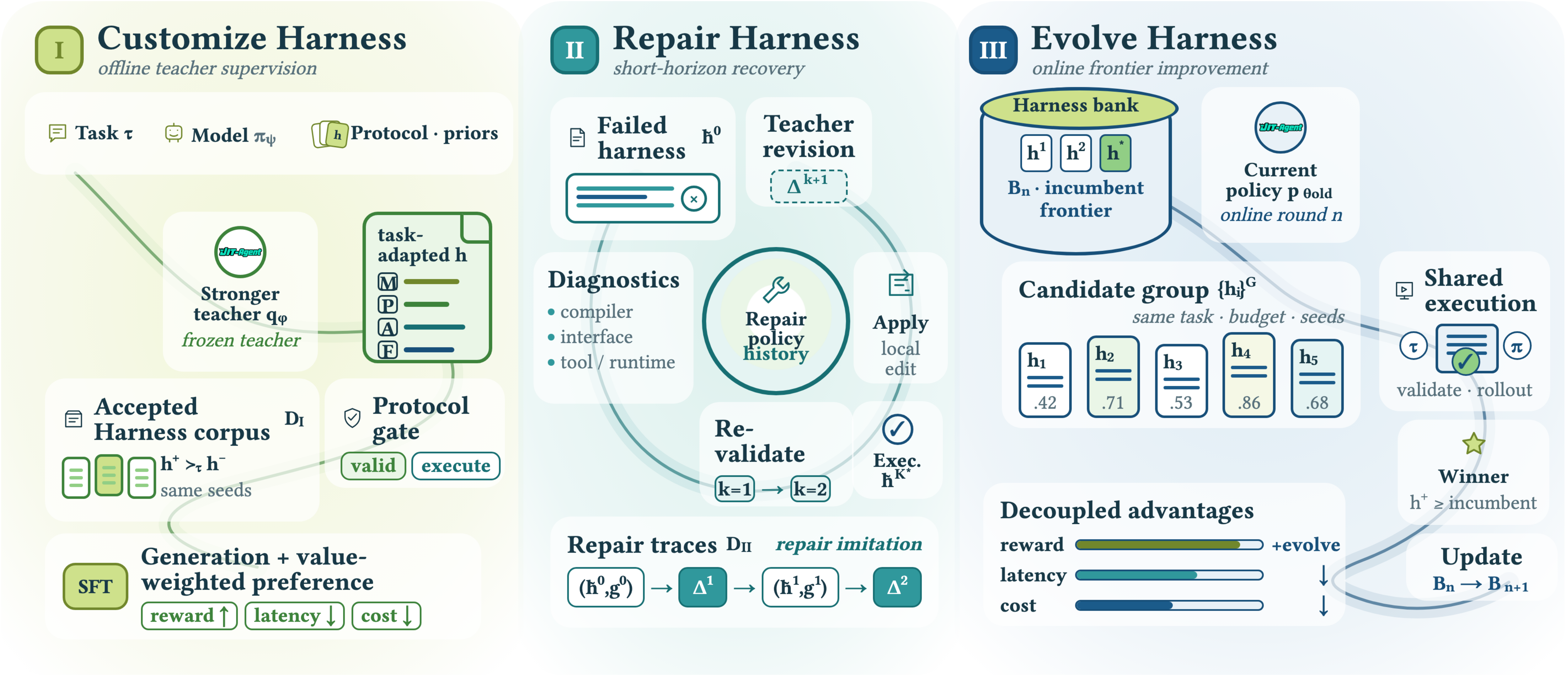

I
Customize Harness
offline teacher supervision
Task τ
Model π_ψ
Protocol · priors
Stronger teacher q_φ
frozen teacher
task-adapted h
M
P
A
F
Accepted Harness corpus D_I
$h^+ \succ_\tau h^-$
same seeds
Protocol gate
valid
execute
SFT
Generation + value-weighted preference
reward ↑
latency ↓
cost ↓
II
Repair Harness
short-horizon recovery
Failed harness $\hbar^0$
Teacher revision
Δ^{k+1}
Diagnostics
• compiler
• interface
• tool / runtime
Repair policy history
Apply local edit
Re-validate
k=1 → k=2
Exec. $\hbar^{K^*}$
Repair traces D_{II}
repair imitation
$(\hbar^0, g^0)$ → Δ^1 → $(\hbar^1, g^1)$ → Δ^2
III
Evolve Harness
online frontier improvement
Harness bank
h^1 h^2 h^*
B_n · incumbent frontier
Current policy $p_{\theta old}$
online round n
Candidate group $\{h_i\}^G$
same task · budget · seeds
h_1 .42
h_2 .71
h_3 .53
h_4 .86
h_5 .68
Shared execution
τ
π
validate · rollout
Winner
$h^+ \geq$ incumbent
Decoupled advantages
reward +evolve
latency ↓
cost ↓
Update
$B_n \rightarrow B_{n+1}$